

HW2

1. The European rocket Ariane 5, used to deliver payloads into geostationary transfer orbit (GTO) or low earth orbit (LEO).

- a. Initially, at liftoff of an Ariane 5 launch vehicle, two P230 solid-propellant boosters plus the main Vulcain engine are ignited. The two different components of the launch vehicle have the following characteristics:

<i>Engine</i>	<i>Vulcain (main engine)</i>	<i>Booster P230</i>
<i>Type</i>	Liquid Propellant	Solid Propellant
<i>Thrust (kN)</i>	1114	6470
<i>Specific Impulse (s)</i>	430	275

Calculate:

- i. Effective velocity for each engine.
 - ii. Exhaust mass flow rate.
 - iii. Average effective exhaust velocity (main engine and boosters).
 - iv. Average specific impulse (main engine and boosters).
 - b. Calculate the total change in rocket velocity when it is used to deliver payloads into low earth orbit (LEO). Details of masses are given below in the table:

Stage	Two boosters	First stage	Second stage
Propellant mass (ton)	237	170	15
Dry mass (ton) (M-M _p)	268	16.3	4.7
Specific Impulse	Find from a.(iv)		446

Payload mass is 10.5 tons.

2. Find the optimal mass for a three-stage launch vehicle that is required to lift a 5000 kg payload of 10 km/s. For each stage,

Stage 1	Isp = 400s	€ = 0.10
Stage 2	Isp = 350s	€ = 0.15
Stage 3	Isp = 300s	€ = 0.20